


[Advertise with MatWeb!](#)
[REGISTER NOW](#)
Data sheets for over **180,000** metals, plastics, ceramics, and composites.
[HOME](#) • [SEARCH](#) • [TOOLS](#) • [SUPPLIERS](#) • [FOLDERS](#) • [ABOUT US](#) • [FAQ](#) • [LOG IN](#)

 Searches: [Advanced](#) | [Category](#) | [Property](#) | [Metals](#) | [Trade Name](#) | [Manufacturer](#) | [Recently Viewed Materials](#)

PTFE

[SEARCH](#)


High-Performance PTFE Extrusions

[LEARN MORE](#)


Overview of materials for Polytetrafluoroethylene (PTFE), Extruded

Categories: [Polymer](#); [Thermoplastic](#); [Fluoropolymer](#); [Polytetrafluoroethylene \(PTFE\)](#); [Polytetrafluoroethylene \(PTFE\), Extruded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polytetrafluoroethylene (PTFE), Extruded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

 Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

[Printer friendly version](#)
[Download as PDF](#)
[Download to Excel \(requires Excel and Windows\)](#)
[Export data to your CAD/FEA program](#)

 Add to Folder: ☐ My Folder ▼ 0/0

Physical Properties	Metric	English	Comments
Density	2.00 - 2.30 g/cc	0.0723 - 0.0831 lb/in ³	Average value: 2.17 g/cc Grade Count:61
Water Absorption	0.000 - 0.0100 %	0.000 - 0.0100 %	Average value: 0.00967 % Grade Count:30
Particle Size	400 - 700 µm	400 - 700 µm	Average value: 504 µm Grade Count:41
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	58.0	58.0	Average value: 58.0 Grade Count:14
Hardness, Shore A	98.0	98.0	Average value: 98.0 Grade Count:14
Hardness, Shore D	50.0 - 67.0	50.0 - 67.0	Average value: 58.2 Grade Count:35
Tensile Strength, Ultimate	13.0 - 55.0 MPa	1890 - 7980 psi	Average value: 25.6 MPa Grade Count:58
Tensile Strength, Yield	20.0 - 196 MPa	2900 - 28400 psi	Average value: 131 MPa Grade Count:3
Elongation at Break	70.0 - 600 %	70.0 - 600 %	Average value: 336 % Grade Count:61
Modulus of Elasticity	0.390 - 0.750 GPa	56.6 - 109 ksi	Average value: 0.599 GPa Grade Count:3
Flexural Modulus	0.275 - 0.700 GPa	39.9 - 102 ksi	Average value: 0.500 GPa Grade Count:30
Compressive Yield Strength	2.00 - 24.1 MPa	290 - 3500 psi	Average value: 7.31 MPa Grade Count:10
Flex Crack Resistance	1.00e+6	1.00e+6	Average value: 1.00e+6 Grade Count:15
Poissons	0.460	0.460	Average value: 0.460 Grade Count:14

Ratio			
Izod Impact, Notched	1.87 - 1.89 J/cm	3.50 - 3.54 ft-lb/in	Average value: 1.88 J/cm Grade Count:29
Coefficient of Friction	0.0200 - 0.200	0.0200 - 0.200	Average value: 0.0929 Grade Count:39
Coefficient of Friction, Static	0.0500 - 0.0800	0.0500 - 0.0800	Average value: 0.0650 Grade Count:14
K (wear) Factor	10.0 - 5040 x 10 ⁻⁸ mm ³ /N-M	4.96 - 2500 x 10 ⁻¹⁰ in ³ -min/ft-lb-hr	Average value: 4080 x 10 ⁻⁸ mm ³ /N-M Grade Count:18
Limiting Pressure Velocity	0.0420 - 0.0876 MPa-m/sec	1200 - 2500 psi-ft/min	Average value: 0.0642 MPa-m/sec Grade Count:14
Taber Abrasion, mg/1000 Cycles	8.90	8.90	Average value: 8.90 Grade Count:14

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00e+14 - 1.00e+19 ohm-cm	1.00e+14 - 1.00e+19 ohm-cm	Average value: 9.00e+17 ohm-cm Grade Count:37
Surface Resistance	1.00e+15 - 1.00e+18 ohm	1.00e+15 - 1.00e+18 ohm	Average value: 8.17e+17 ohm Grade Count:35
Dielectric Constant	2.00 - 2.10	2.00 - 2.10	Average value: 2.09 Grade Count:31
Dielectric Strength	7.44 - 60.0 kV/mm	189 - 1520 kV/in	Average value: 22.1 kV/mm Grade Count:36
Dissipation Factor	0.000100 - 0.000200	0.000100 - 0.000200	Average value: 0.000107 Grade Count:29
Arc Resistance	300 sec	300 sec	Average value: 300 sec Grade Count:28

Thermal Properties	Metric	English	Comments
CTE, linear	75.0 - 790 µm/m-°C	41.7 - 439 µin/in-°F	Average value: 294 µm/m-°C Grade Count:20
Specific Heat Capacity	1.50 J/g-°C	0.359 BTU/lb-°F	Average value: 1.50 J/g-°C Grade Count:14
	1.20 - 1.40 J/g-°C @Temperature 40.0 - 260 °C	0.287 - 0.335 BTU/lb-°F @Temperature 104 - 500 °F	Average value: 1.30 J/g-°C Grade Count:14
Thermal Conductivity	0.167 - 0.300 W/m-K	1.16 - 2.08 BTU-in/hr-ft ² -°F	Average value: 0.292 W/m-K Grade Count:16
Melting Point	321 - 344 °C	610 - 651 °F	Average value: 331 °C Grade Count:32
Maximum Service Temperature, Air	260 - 300 °C	500 - 572 °F	Average value: 261 °C Grade Count:27
Minimum Service Temperature, Air	-270 - -79.0 °C	-454 - -110 °F	Average value: -183 °C Grade Count:26
Flammability, UL94	V-0	V-0	Grade Count:31
Oxygen Index	95.0 %	95.0 %	Average value: 95.0 % Grade Count:30

Processing Properties	Metric	English	Comments
Processing Temperature	379 - 429 °C	714 - 804 °F	Average value: 404 °C Grade Count:13

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.

Users viewing this material also viewed the following:

[Overview of materials for Polyetheretherketone, Unreinforced](#)
[Overview of materials for Nylon 66, Unreinforced](#)
[Overview of materials for Polytetrafluoroethylene \(PTFE\), Molded](#)
[Overview of materials for Fluorinated Ethylene Propylene \(FEP\), Molded/Extruded](#)
[Overview of materials for Polyperfluoroalkoxyethylene \(PFA\), Molded/Extruded](#)

O1902 / 77907

**[Subscribe to Premium Services](#)**

Searches: [Advanced](#) • [Composition](#) • [Property](#) • [Material Type](#) • [Manufacturer](#) • [Trade Name](#) • [UNS Number](#)
Other Links: [Advertising](#) • [Submit Data](#) • [Database Licensing](#) • [Web Design & Hosting](#) • [Trade Publications](#)
[Supplier List](#) • [Unit Converter](#) • [Reference](#) • [Links](#) • [Help](#) • [Contact Us](#) • [Site Map](#) • [FAQ](#) • [Home](#)

[Follow @MatWeb](#)

Please read our [License Agreement](#) regarding materials data and our [Privacy Policy](#). Questions or comments about MatWeb? Please contact us at webmaster@matweb.com. We appreciate your input.

The contents of this web site, the MatWeb logo, and "MatWeb" are Copyright 1996-2024 by MatWeb, LLC. MatWeb is intended for personal, non-commercial use. The contents, results, and technical data from this site may not be reproduced either electronically, photographically or substantively without permission from MatWeb, LLC.